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Lab 9

Question: Demonstrate INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN using SQLite. Create Students and Departments tables and display the results of each join operation.

```
import sqlite3

conn = sqlite3.connect("college.db")

cur = conn.cursor()

cur.execute("Drop table if exists Students")

cur.execute("Drop table if exists Departments")

cur.execute("""Create table Students (student_id INTEGER
    PRIMARY KEY,

            student_name TEXT,

            department_id INTEGER) """)

cur.execute("""

CREATE TABLE Departments (

    department_id INTEGER PRIMARY KEY,

    department_name TEXT

)

""")

cur.executemany("insert into Students values (?,?,?)",

                [(1,"Sanjeev",22),(2,"Mohan",86),

                (3,"Rohan",87),(4,"Ram",None)])

cur.executemany("INSERT INTO Departments VALUES (?, ?)", [

    (22, "CSE"), (86, "ECE"), (88, "Mechanical")

])
```

```
40
41  queries = {
42
43      "INNER JOIN": """
44
45          SELECT s.student_id, s.student_name, d.
                department_name
46
47          FROM Students s INNER JOIN Departments d
48
49          ON s.department_id = d.department_id
50
51      """,
52
53      "LEFT JOIN": """
54
55          SELECT student_id, student_name, department_name
56
57          FROM Students LEFT JOIN Departments
58
59          ON Students.department_id = Departments.
                department_id
60
61      """,
62
63      "RIGHT JOIN": """
64
65          SELECT student_id, student_name, department_name
66
67          FROM Students RIGHT JOIN Departments
68
69          ON Students.department_id = Departments.
                department_id
70
71      """,
72
73      "FULL OUTER JOIN": """
74
75          SELECT student_id, student_name, department_name
76
77          FROM Students FULL OUTER JOIN Departments
78
79          ON Students.department_id = Departments.
                department_id
80
81      """
82
83  }
84
85  for name, query in queries.items():
```

```
86
87      print(f"\n{name}")
88
89      cur.execute(query)
90
91      print("Student ID | Name | Department")
92
93      for row in cur.fetchall():
94
95          print(row)
96
97  conn.close()
```

Output:

```
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab9> python .\
    join.py

INNER JOIN
Student ID | Name | Department
(1, 'Sanjeev', 'CSE')
(2, 'Mohan', 'ECE')

LEFT JOIN
Student ID | Name | Department
(1, 'Sanjeev', 'CSE')
(2, 'Mohan', 'ECE')
(3, 'Rohan', None)
(4, 'Ram', None)

RIGHT JOIN
Student ID | Name | Department
(1, 'Sanjeev', 'CSE')
(2, 'Mohan', 'ECE')
(None, None, 'Mechanical')

FULL OUTER JOIN
Student ID | Name | Department
(1, 'Sanjeev', 'CSE')
(2, 'Mohan', 'ECE')
(3, 'Rohan', None)
(4, 'Ram', None)
(None, None, 'Mechanical')
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab9>
```